

1 MORE INDUSTRIAL BASELINES

Trace	AWS Replication	GCS Replication	Azure Replication
T15	6.0	6.5	8.3
T29	3.0	3.8	4.9
T65	0.98	1.3	1.6
T78	2.0	2.8	3.5
T79	3.0	3.7	4.2

Table 1: Cost comparison of industrial baselines (AWS, GCS, Azure) vs. GlobalStore across traces. Values represent the cost multiplier relative to GlobalStore.

To evaluate GlobalStore against industrial baselines, we conducted experiments comparing its cost-effectiveness to commercial solutions, including Google Cloud Storage (GCS) multi-region replication [2] and Azure Geo-Redundant Storage (with read access) [1]. These baselines represent widely available storage services that offer replication functionality, similar to AWS multi-region bucket. However, both GCS and Azure replication use a primitive replicate-on-write mechanism and incur additional storage costs for replicated regions.

We performed experiments on two US regions (US East and US West) for these baselines, as they do not support multi-cloud setups. We used the same five experimental traces as in Figure 5 and analyzed the overall cost trends. While these baselines differ slightly in pricing and service fees across providers, the overall trend remains consistent, with replicated storage incurring significantly higher costs than GlobalStore. As shown in Table 1, GCS replication costs 1.3 – 6.5 $\times$ higher cost, and Azure replication costs 1.6 – 8.3 $\times$ more than GlobalStore. Azure replication incurs additional service charges on reading the replica in the replicated region. On average, GlobalStore achieves cost efficiency while maintaining consistency guarantees across multiple regions, highlighting its advantage over commercial baselines that focus on single-provider setups.

2 GLOBALSTORE USAGE

```

1
2 # start metadata server
3 skystore start-server
4 # register "skybucket" metadata
5 skystore register --config register_config.json
6
7 # start proxy
8 skystore start-proxy --host localhost --port 8002
9
10 # List operation
11 aws s3api --endpoint-url http://127.0.0.1:8002
12     ↪ list-buckets
13
14 # Put operation

```

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```

14 aws s3api put-object --bucket skybucket --key
    ↪ readme-2 --body "./README.md" --endpoint-
    ↪ url http://127.0.0.1:8002

```

Above is a simple pipeline on how to start metadata server, register logical bucket (with necessary information from the physical bucket), start the S3 client-side proxy, and directly use AWS APIs to interact with GlobalStore. In the register configuration, users can specify a list of available regions (and potentially the read and write access to each region), the policy to use on the server side (e.g., TTL-based, or default to GlobalStore policy), or any physical buckets the user wants to register at GlobalStore.

REFERENCES

- [1] Azure Blob Storage | Microsoft Azure — [azure.microsoft.com](https://azure.microsoft.com/en-us/products/storage/blobs). <https://azure.microsoft.com/en-us/products/storage/blobs>, 2024. [Accessed 20-04-2024].
- [2] Gcp multi-region bucket. <https://cloud.google.com/storage/docs/locations#location-mr>. Accessed on 12/15/2022.